

Internal Cost Analysis

A self-contained explanation of the saved material, labor, machine, outside-processing, and additional costs used to calculate this part price.

Part	ABC-123	Revision	B
Description	Billet adapter <production>	Cost quantity	100
Cost date	2026-08-03 12:30	Reference	Costing run 42
Prepared by	report-user	Current saved cost	Yes

Cost conclusion

PRODUCTION QTY	100	RAW COST/UNIT	\$11.4900	CALCULATED PRICE/UNIT	\$13.5005
Cost element	Raw cost	% raw	Effective factor	Calculated price	Price/unit
Materials	\$294.00	25.6%	1.200x	\$352.80	\$3.5280
Machine	\$405.00	35.2%	1.156x	\$468.00	\$4.6800
Labor	\$335.00	29.2%	1.155x	\$387.00	\$3.8700
Outside processing	\$105.00	9.1%	1.240x	\$130.25	\$1.3025
Additional	\$10.00	0.9%	1.200x	\$12.00	\$0.1200
Total	\$1,149.00	100.0%	1.175x	\$1,350.05	\$13.5005

The ERP retail reference saved with this run was \$8.9900 per unit. It is shown as a comparison only and is not used in the cost build-up above.

Costed line summary

Materials (3)

Material	Qty/part	Raw cost	Calculated price
AL-6061-1 - Aluminum round bar	1.25	\$180.25	\$216.30
SUB-ASSY-20 - Machined valve subassembly	1.25	\$180.00	\$216.00
PACK-01 - Protective packaging set	1.25	\$43.75	\$52.50
Material lines		\$404.00	\$484.80

Operations (5)

Operation	Work center	Raw cost	Calculated price
10 / OP20 - Machine profile and finish bores	MILL	\$165.00	\$189.75
20 / OP20 - Machine profile and finish bores	MILL	\$165.00	\$189.75
30 / ANODIZE - Black hard anodize and seal	OUTSIDE	\$85.00	\$106.25
40 / OP20 - Machine profile and finish bores	MILL	\$165.00	\$189.75
50 / OP20 - Machine profile and finish bores	MILL	\$165.00	\$189.75
Operation lines		\$745.00	\$865.25

Material cost build-up

Backflushed materials only. Source, quantity extension, and cost contribution follow for each line.

1. AL-6061-1 - Aluminum round bar

Source: PO PO-1001-1, 2026-07-15 | \$11.5360/BAR x 0.125 conversion = \$1.4420/EA.

Question	Calculation	Result
Net quantity required	1.25 per assembly x 100 assemblies	125
Purchase increment	Saved minimum/increment used for rounding	25
Quantity charged	Net required 125 + rounding/waste 0	125

Line cost: 125 x \$1.4420 = \$180.25 raw; x 1.200x = \$216.30.

Cost component	Raw cost	Factor	Calculated price
Material	\$180.25	1.200x	\$216.30
Line contribution	\$180.25	1.200x	\$216.30

2. SUB-ASSY-20 - Machined valve subassembly

Source: manufactured cost run 317 (current) | \$1.4400/unit; component cost buckets carried through below.

Question	Calculation	Result
Net quantity required	1.25 per assembly x 100 assemblies	125
Purchase increment	Saved minimum/increment used for rounding	25
Quantity charged	Net required 125 + rounding/waste 0	125

Line cost: 125 x \$1.4400 = \$180.00 raw; x 1.200x = \$216.00.

Cost component	Raw cost	Factor	Calculated price
Material	\$70.00	1.200x	\$84.00
Machine inherited from component	\$45.00	1.200x	\$54.00
Labor inherited from component	\$35.00	1.200x	\$42.00
Outside processing inherited from component	\$20.00	1.200x	\$24.00
Additional inherited from component	\$10.00	1.200x	\$12.00
Line contribution	\$180.00	1.200x	\$216.00

3. PACK-01 - Protective packaging set

Source: manual override | \$0.3500/unit.

Question	Calculation	Result
Net quantity required	1.25 per assembly x 100 assemblies	125
Purchase increment	Saved minimum/increment used for rounding	25
Quantity charged	Net required 125 + rounding/waste 0	125

Line cost: 125 x \$0.3500 = \$43.75 raw; x 1.200x = \$52.50.

Cost component	Raw cost	Factor	Calculated price
Material	\$43.75	1.200x	\$52.50
Line contribution	\$43.75	1.200x	\$52.50

Operation cost build-up

Saved routing sequence with time and rate equations for internal work and PO pricing for outside work.

10. OP20 - Machine profile and finish bores

Work center: MILL | Demand: 1/part x 100 parts = 100 units.

Driver	Calculation	Result
Batches	Ceiling(100 processed units / 25 batch size)	4
Batch resets	Maximum(4 batches - 1, 0)	3
Running time	100 units x 0.08 cycle hr/unit	8 hr
Occupied time	1.5 setup + 8 running + 3 x (0.1 reset + 0.05 idle)	9.95 hr
After-hours idle	Saved scheduling result	0 hr

Labor: (1.5 setup hr x \$25.00/hr) + (3 resets x 0.1 hr x \$25.00/hr) = \$75.00 raw.

Machine: (9.95 occupied hr x \$5.00/hr) + (0 after-hours hr x \$5.00/hr x 1) + (8 running hr x \$10.00/hr) = \$90.00 raw.

Cost component	Raw cost	Factor	Calculated price
Machine	\$90.00	1.150x	\$103.50
Labor	\$75.00	1.150x	\$86.25
Operation contribution	\$165.00	1.150x	\$189.75

20. OP20 - Machine profile and finish bores

Work center: MILL | Demand: 1/part x 100 parts = 100 units.

Driver	Calculation	Result
Batches	Ceiling(100 processed units / 25 batch size)	4
Batch resets	Maximum(4 batches - 1, 0)	3
Running time	100 units x 0.08 cycle hr/unit	8 hr
Occupied time	1.5 setup + 8 running + 3 x (0.1 reset + 0.05 idle)	9.95 hr
After-hours idle	Saved scheduling result	0 hr

Labor: (1.5 setup hr x \$25.00/hr) + (3 resets x 0.1 hr x \$25.00/hr) = \$75.00 raw.

Machine: (9.95 occupied hr x \$5.00/hr) + (0 after-hours hr x \$5.00/hr x 1) + (8 running hr x \$10.00/hr) = \$90.00 raw.

Cost component	Raw cost	Factor	Calculated price
Machine	\$90.00	1.150x	\$103.50
Labor	\$75.00	1.150x	\$86.25
Operation contribution	\$165.00	1.150x	\$189.75

30. ANODIZE - Black hard anodize and seal

Work center: OUTSIDE | Demand: 1/part x 100 parts = 100 units.

Source: PO PO-88214-3, 2026-07-22 | 100 units x \$0.8500 = \$85.00 raw; x 1.250x = \$106.25.

Cost component	Raw cost	Factor	Calculated price
Outside processing	\$85.00	1.250x	\$106.25
Operation contribution	\$85.00	1.250x	\$106.25

40. OP20 - Machine profile and finish bores

Work center: MILL | Demand: 1/part x 100 parts = 100 units.

Driver	Calculation	Result
Batches	Ceiling(100 processed units / 25 batch size)	4
Batch resets	Maximum(4 batches - 1, 0)	3
Running time	100 units x 0.08 cycle hr/unit	8 hr
Occupied time	1.5 setup + 8 running + 3 x (0.1 reset + 0.05 idle)	9.95 hr
After-hours idle	Saved scheduling result	0 hr

Labor: (1.5 setup hr x \$25.00/hr) + (3 resets x 0.1 hr x \$25.00/hr) = \$75.00 raw.

Machine: (9.95 occupied hr x \$5.00/hr) + (0 after-hours hr x \$5.00/hr x 1) + (8 running hr x \$10.00/hr) = \$90.00 raw.

Cost component	Raw cost	Factor	Calculated price
Machine	\$90.00	1.150x	\$103.50
Labor	\$75.00	1.150x	\$86.25
Operation contribution	\$165.00	1.150x	\$189.75

50. OP20 - Machine profile and finish bores

Work center: MILL | Demand: 1/part x 100 parts = 100 units.

Driver	Calculation	Result
Batches	Ceiling(100 processed units / 25 batch size)	4
Batch resets	Maximum(4 batches - 1, 0)	3
Running time	100 units x 0.08 cycle hr/unit	8 hr
Occupied time	1.5 setup + 8 running + 3 x (0.1 reset + 0.05 idle)	9.95 hr
After-hours idle	Saved scheduling result	0 hr

Labor: (1.5 setup hr x \$25.00/hr) + (3 resets x 0.1 hr x \$25.00/hr) = \$75.00 raw.

Machine: (9.95 occupied hr x \$5.00/hr) + (0 after-hours hr x \$5.00/hr x 1) + (8 running hr x \$10.00/hr) = \$90.00 raw.

Cost component	Raw cost	Factor	Calculated price
Machine	\$90.00	1.150x	\$103.50
Labor	\$75.00	1.150x	\$86.25
Operation contribution	\$165.00	1.150x	\$189.75

Reconciliation and review flags

This audit ties the saved header totals back to the saved material and operation lines. Non-zero variance indicates an inconsistent saved snapshot that should not be used for quoting.

Bucket	Header raw	Lines raw	Variance	Header price	Lines price	Variance
Materials	\$294.00	\$294.00	\$0.00	\$352.80	\$352.80	\$0.00
Machine	\$405.00	\$405.00	\$0.00	\$468.00	\$468.00	\$0.00
Labor	\$335.00	\$335.00	\$0.00	\$387.00	\$387.00	\$0.00
Outside processing	\$105.00	\$105.00	\$0.00	\$130.25	\$130.25	\$0.00
Additional	\$10.00	\$10.00	\$0.00	\$12.00	\$12.00	\$0.00
Total	\$1,149.00	\$1,149.00	\$0.00	\$1,350.05	\$1,350.05	\$0.00

Review flags

- PACK-01: uses a manual unit-cost override.

Internal notes

Review setup assumptions.

Approved for quoting.